

WELCOME TO INDUSTRY 4.0

Industry 4.0 represents the next phase of industrial transformation where physical systems, digital technologies and intelligent automation work together to create **smart and connected enterprises**.

SESSION 1 AGENDA

- | | | |
|----|---|-------------------------------------|
| 01 |  | Welcome to Industry 4.0 |
| 02 |  | Evolution of Industrial Revolutions |
| 03 |  | What is Industry 4.0? |
| 04 |  | Why Industry 4.0 Matters |
| 05 |  | Key Enabling Technologies |
| 06 |  | Industry 4.0 in Action |



“ From automated machines
to intelligent ecosystems. ”



EVOLUTION OF INDUSTRY

A journey of innovation that has transformed the way we live and work



Personal smart gadgets

Health trackers

Smart homes,

Smart factories

Data-driven healthcare

Intelligent infrastructure

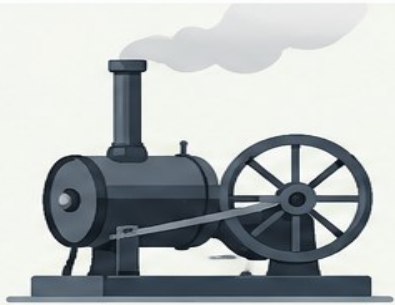
Automated traffic systems

Drone delivery networks.

Industry 4.0 – Connected & Intelligent systems

EVOLUTION OF INDUSTRY

A journey of innovation that has transformed the way we live and work



INDUSTRY 1.0

 Late 1700s

Steam Power

Mechanization of production with water and steam power

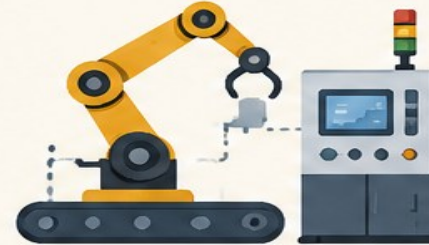


INDUSTRY 2.0

 Early 1900s

Electricity & Mass Production

Introduction of electricity enabling mass production and assembly lines



INDUSTRY 3.0

 1970s Onwards

Computers & Automation

Use of electronics, IT and automation to improve efficiency and quality



INDUSTRY 4.0

 Today

Connected Intelligent Systems

Integration of cyber-physical systems, IoT, AI and data driving smart operations



Key Message: Each industrial revolution introduced a new generation of technology, driving higher productivity, better quality and improved human life.



WHAT IS INDUSTRY 4.0?

Industry 4.0 is the integration of physical systems, digital technologies, and intelligent automation to create **smart, connected and agile enterprises**.



KEY BENEFITS



Higher Efficiency

Optimize operations and reduce waste



Better Quality

Improve quality with real-time monitoring and analytics



Reduced Downtime

Predict issues and prevent equipment failures



Faster Decision Making

Access real-time insights for quick actions



Lower Costs

Improve productivity and reduce operational costs



INDUSTRY 4.0 FORMULA



IoT

+



AI

+



Cloud

+



Data Analytics

+



Automation

+



Cybersecurity

Working Together to Create Smart, Connected and Future-Ready Enterprises

WHAT IS INDUSTRY 4.0?

Industry 4.0 is the integration of physical systems, digital technologies, and intelligent automation to create **smart, connected and agile enterprises**.



Physical
Systems

Digital
Technologies

Intelligent
Automation

Machines equipped with sensors

Generates data

Data is transmitted through networks,

AI transforms data into actionable insights

Systems automatically operates, predicts & optimizes

WHAT IS INDUSTRY 4.0?

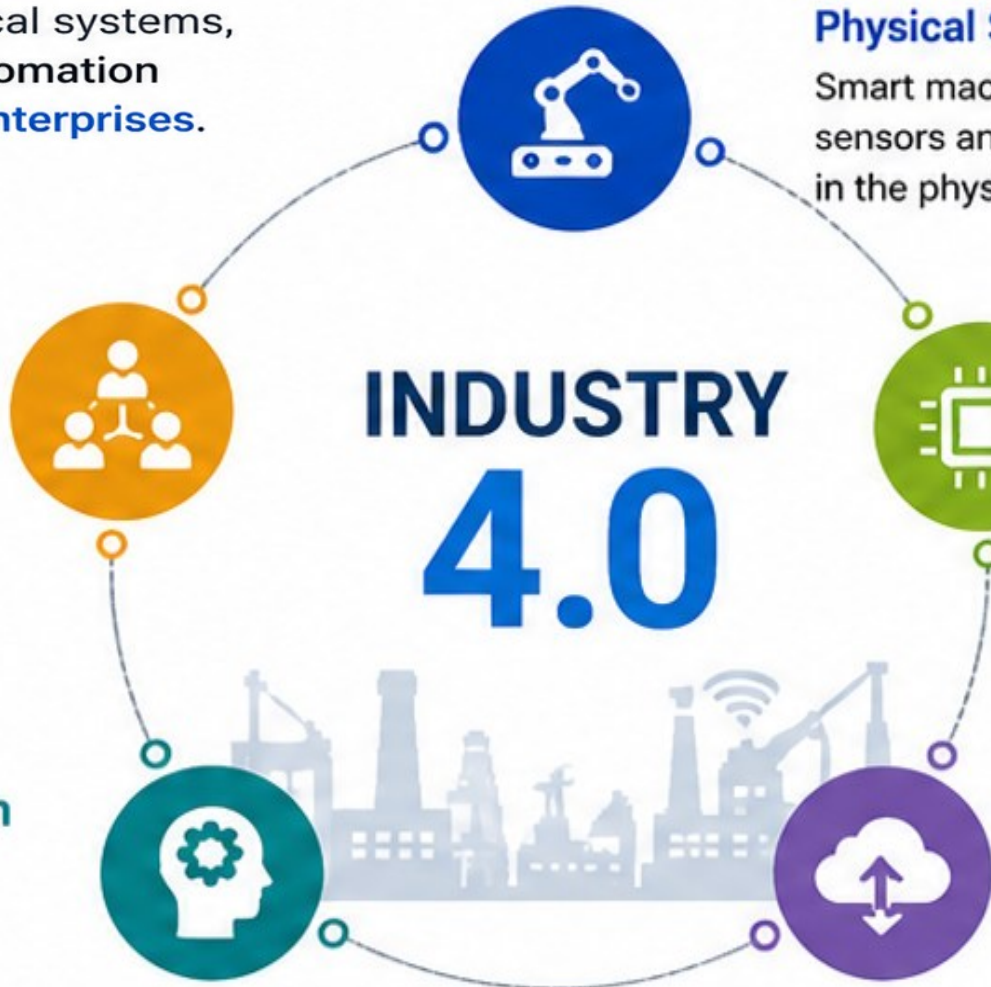
Industry 4.0 is the integration of physical systems, digital technologies, and intelligent automation to create **smart, connected and agile enterprises**.

Connected Ecosystems

People, processes, systems and machines work together

Intelligent Automation

AI and automation enable adaptive and autonomous actions



Physical Systems

Smart machines, sensors and devices in the physical world

Digital Technologies

IoT, AI, Cloud, Big Data and advanced analytics

Real-Time Data

Continuous data collection, processing and insights

WHAT IS INDUSTRY 4.0?

Industry 4.0 is the integration of physical systems, digital technologies, and intelligent automation to create **smart, connected and agile enterprises**.



KEY BENEFITS



Higher Efficiency

Optimize operations and reduce waste



Better Quality

Improve quality with real-time monitoring and analytics



Reduced Downtime

Predict issues and prevent equipment failures



Faster Decision Making

Access real-time insights for quick actions



Lower Costs

Improve productivity and reduce operational costs



INDUSTRY 4.0 FORMULA



IoT

+



AI

+



Cloud

+



Data Analytics

+



Automation

+



Cybersecurity

Working Together to Create Smart, Connected and Future-Ready Enterprises

WHY INDUSTRY 4.0 MATTERS

Organizations are under pressure to do more with less.
Industry 4.0 helps turn these challenges into opportunities.

TODAY'S BUSINESS CHALLENGES

-  **Increasing Competition**
Global competition is stronger than ever.
-  **Pressure to Reduce Costs**
Need to lower operational costs while improving efficiency.
-  **Changing Customer Expectations**
Customers expect high quality, personalization and fast delivery.
-  **Unplanned Downtime**
Equipment failures lead to delays, higher costs and unhappy customers.
-  **Complex Supply Chains**
Lack of visibility and coordination creates delays and risks.
-  **Need for Sustainability**
Organizations must use resources responsibly and reduce waste.



HOW INDUSTRY 4.0 HELPS

-  **Improve Productivity**
Automate processes and optimize resource utilization.
-  **Enhance Quality**
Use real-time monitoring and predictive insights to ensure consistent quality.
-  **Reduce Downtime**
Predict equipment failures and schedule maintenance proactively.
-  **Lower Operational Costs**
Optimize operations and reduce waste across the value chain.
-  **Gain End-to-End Visibility**
Track assets, inventory and shipments in real time.
-  **Achieve Sustainability Goals**
Optimize energy usage and reduce environmental impact.

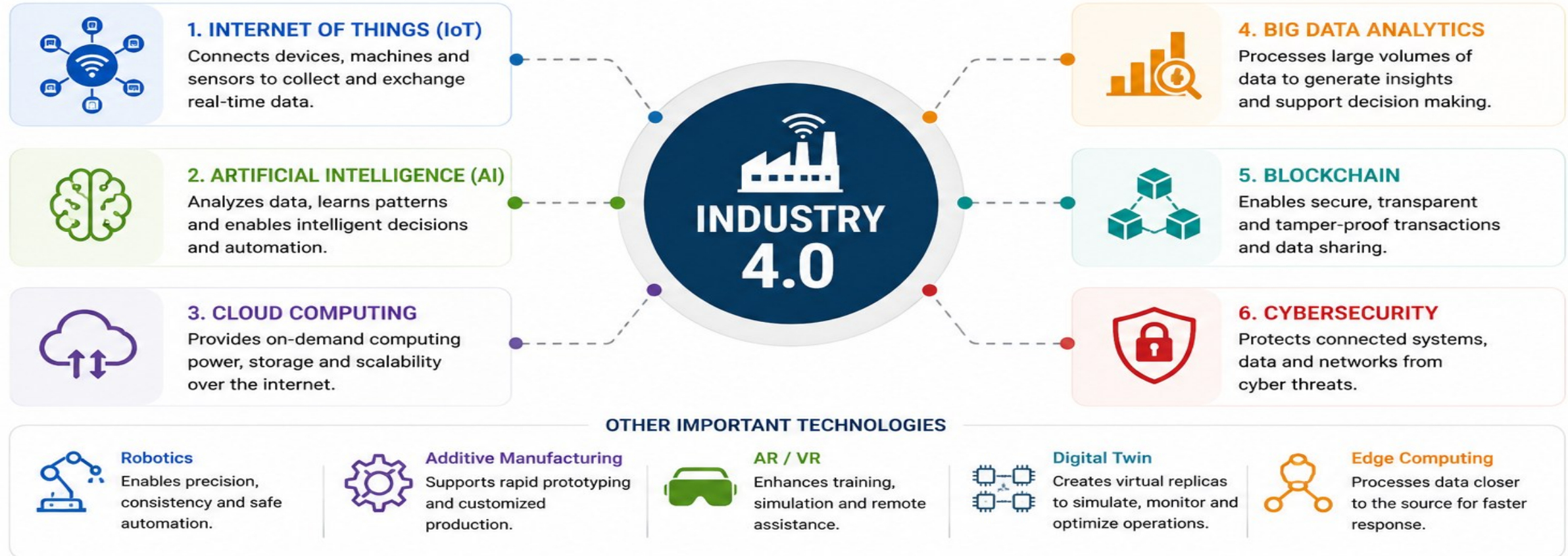


Industry 4.0 is not just about technology – it's about transforming the way we work, creating value, and building a better future.



KEY ENABLING TECHNOLOGIES OF INDUSTRY 4.0

Industry 4.0 is powered by a set of advanced technologies working together to create intelligent, connected and future-ready enterprises.



WORKING TOGETHER

These technologies work in synergy to build smart factories, connected supply chains, intelligent products and data-driven business models.



INDUSTRY 4.0 IN ACTION

Transforming industries and creating value across the world

REAL-WORLD APPLICATIONS ACROSS KEY INDUSTRIES



MANUFACTURING

- Smart factories with connected machines and sensors
- Predictive maintenance reduces downtime and boosts productivity



SUPPLY CHAIN & LOGISTICS

- Real-time tracking of goods and shipments
- Better demand forecasting and inventory optimization



HEALTHCARE

- Connected medical devices and remote patient monitoring
- Data-driven insights for better diagnosis and treatment



ENERGY & UTILITIES

- Smart grids and meters for efficient energy management
- Predictive analytics for equipment health and outage management



AUTOMOTIVE

- Smart manufacturing and robotic automation
- Connected vehicles for real-time diagnostics and safety



RETAIL

- Personalized customer experiences
- Smart inventory and omnichannel operations



BANKING & FINANCE

- Secure digital transactions with blockchain
- AI-powered fraud detection and risk management



TRANSPORTATION

- Intelligent traffic management and route optimization
- Predictive maintenance for vehicles and infrastructure



AGRICULTURE

- IoT sensors for soil, weather and crop monitoring
- Data-driven decisions to improve yield and reduce waste



KEY TAKEAWAY

Industry 4.0 is not limited to one sector. It is driving innovation, improving efficiency, enhancing customer experiences and creating sustainable value across every industry.



KEY TAKEAWAYS

Session 1: Introduction to Industry 4.0



WHAT WE LEARNED

- ✓ Industry 4.0 is the **Fourth Industrial Revolution**.
- ✓ It combines **physical** and **digital** worlds.
- ✓ Multiple technologies work together to create **intelligent enterprises**.
- ✓ **IoT, AI, Cloud, Analytics, Blockchain** and **Cybersecurity** are key enablers.
- ✓ Industry 4.0 is transforming **every industry**, not just manufacturing.



FOOD FOR THOUGHT

Imagine a factory where machines can monitor themselves, predict failures, order spare parts automatically, and optimize production without human intervention.

— What technologies —
would make this possible?



LOOKING AHEAD

Next Session:
Core Pillars of Industry 4.0

Deep Dive Into:



IoT



AI & Machine
Learning



Cloud
Computing



Big Data
Analytics



Blockchain



Digital
Twins



Edge
Computing



You have taken the first step toward understanding
the future of Industry. Let's build it together!

